Bishnu Prasad Kar

AI/ML Engineer – Machine Learning, LLMs, Agentic AI, Quantum Computing

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ABOUT

AI/ML engineer and Computer Science undergraduate (B.Tech, expected August 2027) who builds complete systems and measures whether they work. Recent work spans a lab-report assistant whose OCR pipeline, statistics and LLM explanations are each evaluated and gated, and a relativistic universe simulator in Rust with every physics model tested against observation. Works in Python, Rust and TypeScript, from model training and retrieval to APIs, databases, GPU code and CI. Three conference papers presented (2024–2026); results are reported against baselines, with their limits stated plainly.

Research interests: Reliable and verifiable machine learning; LLM reasoning, evaluation and guardrails; retrieval-grounded generation; medical document understanding and image segmentation; scientific computing and simulation; quantum machine learning (QAOA, quantum kernels).

EDUCATION

Qualification	Institution	Year
B.Tech, Computer Science and Engineering	Gandhi Institute of Excellent Technocrats, Bhubaneswar	Expected 2027
Higher Secondary (Class 12), Science	R.H.S.S, Mancheswar, Bhubaneswar	2020 - 2022
Secondary (Class 10)	S.S.V.M. Unit-3, Bhubaneswar	2020

CGPA 8.8/10 through semester 6 (currently semester 7). Coursework: Machine Learning, Artificial Intelligence, Data Structures and Algorithms, Database Management Systems, Operating Systems, Computer Networks, Digital Logic Design.

INTERNSHIPS AND TRAINING

Data Science Intern Dec 2025 - Feb 2026

Jyesta Corporate Entity x E-Cell, IIT Roorkee

- Built recommendation and loan-prediction systems with feature engineering, several classifiers and explainability.

Trainee: Generative AI, Data Analytics, Cloud Computing 2024 - 2026

Central Tool Room and Training Centre (CTTC), Ministry of MSME

- Generative AI (2026): LLMs, Hugging Face, LangChain, RAG, end-to-end capstone. Data Analytics (2025): Python, SQL, Tableau.

Backend Engineering Trainee Jul 2025 - Jul 2026

Boot.dev (remote)

- Full backend track: Python, SQL, Linux, Docker, networking, system design.

MERN Stack Developer Intern Apr 2025 - May 2025

Codec Technologies India

- One-month project internship building full-stack web applications.

PROJECTS

Nabz – Lab Reports Explained Aug – Oct 2026

Python, FastAPI, PostgreSQL 17, pgvector, RapidOCR, Groq, React, TypeScript, Three.js, Docker

github.com/Bishnu2430/Nabz | **Complete**

- Reads a photo or PDF of a blood-test report, has the user confirm every value, then explains the results in plain English, Hindi or Odia, maps them onto an interactive 3D body and tracks them across years. Built to help people prepare for their doctor; it does not diagnose.
- Extraction: the PDF text layer first, else RapidOCR (ONNX, CPU) after projection-profile deskew, then a layout-agnostic row parser, LOINC matching over a 108-test catalogue, unit normalisation and a trained row-confidence model that sends weak rows to review first. 100% field accuracy on 2,228 rows from digital PDFs; 94.6% row recall at 100% precision on simulated phone photos.
- Analysis: lab-range classification, fixed critical-value alerts, reference change values, Theil–Sen trends with an exact Mann–Kendall test, and NHANES population percentiles. On 50 synthetic histories it confirms 88% of planted drifts at five visits with 1.0% false alarms.
- Explanations: gpt-oss-120b writes to a strict JSON schema from retrieved NLM MedlinePlus passages in pgvector; a deterministic validator (script, numbers, citations, banned intents in three languages), then an LLM judge, then a template fallback. No generated text appears near a critical value; 77 red-team cases run in CI.
- Platform: API and worker queue, 22-table schema with Alembic, accounts with two-step sign-in for staff, doctor sharing by QR link, data export and deletion, an OWASP ASVS 5.0 Level 1 self-assessment, 13 architecture decision records, and CI for backend, frontend and diagrams.

Worldline – Relativistic Universe Sandbox Oct 2026 – present

Rust, wgpu, egui; data from NASA JPL and NAIF

github.com/Bishnu2430/Worldline | **In progress (pre-alpha)**

- Desktop engine that runs the real solar system in 3D: 459 moons, 62 dwarf planets, asteroids and comets, 28,331 belt objects, the solar wind, heliosphere and magnetospheres. Users drop black holes, neutron stars and planets into it and watch general relativity play out.
- Gravity: Einstein–Infeld–Hoffmann relativistic n-body dynamics with an adaptive IAS15 integrator, planets' gravity fields, post-Newtonian binaries with radiation reaction, and motion in exact Kerr spacetime. Earth stays within 0.41 km of NASA JPL after a simulated year (61 km without relativity).
- Every physics model ships with a validation test against theory or observation, 65+ passing: Mercury's relativistic precession 42.9807″ vs. 42.9805″ per century; the Hulse–Taylor pulsar's orbital decay within 1.1σ of the measured rate; a GW150914 replay leaving 63.0 $M_\odot$ with spin 0.68 (LIGO: 62 ± 4, 0.67); Halley's Comet returning 17 minutes from JPL's 2061 time.
- Compact objects from first principles: the Chandrasekhar limit (1.456 $M_\odot$), a 2.048 $M_\odot$ maximum neutron-star mass from the SLy equation of state, and Hawking evaporation.
- GPU work in wgpu: ray-traced shadows of spinning black holes with Doppler-shifted accretion disks, and SPH fluid planets that tides tear apart, 8,144 particles at 32–55 frames a second on integrated graphics.

Q-CONSENSUS – Multi-Agent LLM Reasoning with Verifiable Decision Trails 2026

Solo. Python, FastAPI, llama.cpp, Qiskit Aer, Solidity, Geth, React, Docker

github.com/Bishnu2430/Q-CONSENSUS

- Five agents debate a question on a local LLM served by llama.cpp; a QAOA solver over quantum-kernel similarity between their answers picks the consensus, benchmarked against a classical baseline.
- Every event is SHA-256 hash-chained and anchored by a Solidity contract on a private Ethereum (Geth) chain, so any run can be replayed and verified after the fact.
- FastAPI service with a React front end; pytest suites cover the debate flow, the quantum optimiser, contract anchoring, reconciliation and API security.

Misinformation Vaccine – Fake-News Detection, URL to Verdict 2025

Solo. Python, PyTorch, Hugging Face Transformers, scikit-learn, FastAPI, PostgreSQL

github.com/Bishnu2430/Misinformation-Vaccine

- Fine-tuned DistilBERT on 43,944 news articles: 99.95% accuracy and 0.9995 F1, against a 98.84% TF-IDF and logistic-regression baseline.
- End-to-end pipeline from a URL to a verdict: scraping, a FastAPI inference service and stored history in PostgreSQL.
- Documented the dataset-bias limits behind the near-ceiling scores rather than presenting them as real-world accuracy.

Floodwatch – IoT Flood Early Warning Feb 2026

Team; 1st position, Model Fiesta, Suravi 2K26. Python, scikit-learn, SHAP, FastAPI, PostgreSQL, Ollama, React, Arduino

github.com/Bishnu2430/Flood_Detection_System

- An Arduino streams water-level, rain and float-switch readings over serial to a FastAPI service, which engineers 14 rolling features and runs a four-level risk model with SHAP attributions.
- At high risk, a local LLM (Ollama) explains the reading and the service switches on a hardware alert on the board; every reading is stored, even when the model or LLM fails.
- React dashboard with live readings, trends, SHAP explanations and streamed LLM output.

QuantumLab – Interactive Quantum Computing Course 2026

Team Creative Cartel, Internal SIH, GIET. Next.js, TypeScript, FastAPI, Qiskit Aer, PostgreSQL, Docker

github.com/Bishnu2430/quantumLab

- Seventeen lessons from a single qubit to Grover's search and BB84, each with worked derivations, an interactive visual, a circuit to run and Python to execute in a sandbox.
- Every circuit runs on Qiskit Aer, and CI checks every stated result against the simulator, so a lesson whose physics does not reproduce fails the build and does not ship.

Soil-to-Shelf – Blockchain Traceability for Ayurvedic Herbs 2025

Team, Smart India Hackathon (internal round), GIET. Hyperledger Fabric

github.com/Bishnu2430/ayurvedic-blockchain

- Permissioned ledger that records geo-tagged collection events at harvest and follows each batch through labs and processors to a QR code on the product, so consumers and regulators can verify provenance.

CycleSense – IoT and LLM Menstrual Health Platform 2025 – 2026

Team; 3rd place, ReFlow hackathon, IIT Bhubaneswar

- Menstrual-health platform pairing IoT sensing with LLM-generated cycle insights.

ShelfSight – Smart Product Scanner 2025

Solo. Python, OpenCV, PaddleOCR, Tesseract, XGBoost, Flask, SQLite

github.com/Bishnu2430/Smart_Product_Scanner

- Generates and scans per-unit QR codes carrying product metadata, reads printed labels with OCR, and serves a demand-prediction model trained on stock-consumption data through an API.

CONFERENCE PAPERS

Presented at and certified by the conferences below; none published in a journal.

J. Dash, **B. P. Kar**, R. L. Mohanty

Entropy-Optimized Adaptive Hybrid Thresholding for Retinal Blood Vessel Segmentation

SPARK-2K26 Research Conclave, GIET Bhubaneswar, January 2026

Neural ODE-UNet manuscript: F1 0.797 and AUC 0.976 on DRIVE.

P. Mohanty, **B. P. Kar**, M. Jena, P. P. Upadhyay

CityFlow: A Scalable Big Data Framework for Real-Time Analytics in Computing 4.0

National Conference on Computing 4.0 (NCCENGT-2025), GIET Bhubaneswar, September 2025

Streaming ingestion, distributed storage and Spark processing, validated on the NYC Taxi dataset.

B. P. Kar, P. Panigrahi, S. N. Mishra

A Modular and Scalable Arithmetic and Logical Unit Design through Multiplexer-Based Controller

International Conference on Semiconductor Technologies and Renewable Energy (ICSTRE-2024), November 2024

8-bit ALU with multiplexer-based operation selection and timing verification.

TECHNICAL SKILLS

Languages	Python, Rust, TypeScript, SQL, Bash
Machine learning	PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face Transformers, NLP, computer vision, OCR (RapidOCR, PaddleOCR, Tesseract), SHAP, model evaluation
Generative AI	LLMs, agentic systems, RAG, LangChain, FAISS, pgvector, llama.cpp, Ollama, Groq and Gemini APIs, structured output, validators and LLM judges, red-team evaluation
Backend, data	FastAPI, Flask, REST APIs, PostgreSQL, MySQL, Alembic, Hadoop, Spark, pandas, NumPy, Tableau
Frontend	React, TypeScript, Next.js, Vite, Tailwind CSS, Three.js
Infrastructure	Docker, Docker Compose, Git, GitHub Actions, Linux
Scientific computing	Numerical integration (IAS15), n-body and SPH simulation, GPU programming with wgpu, ray tracing
Quantum computing	Qiskit, Qiskit Aer, QAOA, quantum kernels, variational circuits
Distributed ledger	Hyperledger Fabric, Solidity, Ethereum (Geth), SHA-256 hash chaining
Hardware, IoT	Arduino, ESP32, sensor integration, serial and MQTT pipelines
Languages spoken	English, Hindi, Odia

AWARDS

- **1st position, Model Fiesta**, Suravi 2K26, GIET Bhubaneswar (Feb 2026) – Floodwatch flood early-warning system.
- **3rd place, ReFlow Menstrual Health Innovation Hackathon**, IIT Bhubaneswar (Nov 2025) – CycleSense.
- **Top performer**, 8-week Machine Learning training, Internshala – 94% final assessment (2025).
- **Completion with Merit**, one-month Quantum Computing course with workshops, C-DAC with IIT Roorkee – 81.67%.

CERTIFICATIONS

AI and deep learning	NVIDIA DLI, Getting Started with Deep Learning (2025); Harvard CS50 AI with Python (2025); IBM SkillsBuild (2025)
Generative AI	Microsoft and LinkedIn, Career Essentials (2024); Google Cloud (2025); CTTC, Ministry of MSME (2026)
ML and data science	Internshala, 8-week ML training, 94% (2025); University of Michigan, Data Science in Python (2024)
Big data	UC San Diego, four-course Big Data specialisation (2024-2025)
Quantum computing	C-DAC with IIT Roorkee, Quantum Computing with Merit, 81.67%
Simulations	Deloitte Data Analytics and KPMG Technology job simulations, Forage (2025)

HACKATHONS AND EVENTS

Hackathons	Internal Smart India Hackathon, GIET (2024, 2025, 2026, Team Creative Cartel); ReFlow, IIT Bhubaneswar (2025); Chakravyuh 1.0 national hackathon (2026); InnovateX 4.0 (2025)
Model Fiesta, Suravi	1st position 2K26; water quality prediction 2K25; participant 2K24
Events	AI Day Bhubaneswar 2025; AWS Summit India 2025; Google Gemini workshop 2025; machine learning workshops (PythonSoft, NSIC and MSME)